

Name _____

Score _____

Answer the problems on separate paper. You do not need to rewrite the problem statements on your answer sheets. Work carefully. Do your own work. **Show all relevant supporting steps!** Attach this sheet to the front of your answers.

1. (8 pts) Use the method of *Reduction of Order* to find a second linearly independent solution y_2 of the differential equation $x^2 y'' - 3xy' + 5y = 0$ given a known solution of $y_1 = x^2 \cos(\ln x)$.
2. (8 pts) Find the general solution of the differential equation $2y'' + 6y' - y = 0$
3. (8 pts) Find the general solution of the differential equation $6y'' - 4y' + y = 0$
4. (14 pts) Solve the initial-value problem $y'' + 8y' + 16y = 0$, $y(0) = 6$, $y'(0) = -2$
5. (14 pts) Find the general solution of the differential equation $y'' - 3y' - 10y = 3 - 5x^2$
6. (14 pts) Find the general solution of the differential equation $y'' - 4y = 6e^{-2x}$
7. (8 pts) Find the general solution of the differential equation $2x^2 y'' - 9xy' + 9y = 0$
8. (8 pts) Find the general solution of the differential equation $2x^2 y'' - 4xy' + 9y = 0$
9. (14 pts) Use the method of *Variation of Parameters* to find a particular solution of $y'' - 2y' + y = e^x \ln x$
10. (10 pts) A mass weighing 9 lbs is attached to a 4 ft spring and stretches the spring so that its new length is 7 ft. This mass is replaced by another mass which weighs 12 lbs. The mass is released at a point $\frac{1}{2}$ ft below the equilibrium position with an initial upward velocity of 1 ft/s. Find the equation of motion.